

Aurchi Chowdhury

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📍 Dhaka, Bangladesh

🌐 Minamotooo

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Interests

Research Interests

Education

Bangladesh University of Engineering and Technology, Computer Science and Engineering

Dhaka, Bangladesh
Nov 2022 – present

Viqarunnisa Noon School and College, High School

- Board Scholarship

Dhaka, Bangladesh
Jan 2009 – Dec 2021

Publications

Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering

San Diego, CA

Aurchi Chowdhury

github.com/azraihaan/mul-vlm-mech-interp

Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception

Denver, CO

Aurchi Chowdhury

github.com/azraihaan/affordance_crystallization

Experience

Progya: Adaptive Tutoring Platform, Dept. of CSE, BUET (UGC-funded), Undergraduate Research Intern

Dhaka, Bangladesh
Nov 2025 – May 2026
7 months

- Developed an adaptive tutoring platform using Bayesian Knowledge Tracing (BKT), building learning pathways aligned with the BCS exam syllabus.
- Supervisors: Md. Saifur Rahman, Md. Shariful Islam Bhuiyan

Projects

MemoryLane

Apr 2026 – present

PyTorch, ConvNeXt, U-Net, PatchGAN, VGG

- Framed grayscale-to-color mapping as a conditional image generation problem in the CIELab color space, building a ConvNeXt-Base encoder – U-Net decoder architecture.
- Incorporated a PatchGAN discriminator and mitigated gray-mean saturation bias via colorfulness-filtered training on ~518K Places365 images.
- Designed a composite perceptual + adversarial loss (VGG feature matching + PatchGAN) to balance global structure with local texture fidelity.
- Surpassed ground-truth colorfulness (40.9 vs. 36.9) and generalized to real, out-of-distribution family photographs.

Scramble

July 2025 – present

ATmega32, SG90 servo motors, HC-05 Bluetooth module, Kotlin

- Developed a Fully Automated Rubik's Cube Solver capable of generating solutions in a maximum of 20 moves.

On the Road

Mar 2024 – present

Node, Express, React, PostgreSQL

- Built a full-stack travel planning platform covering accommodations, dining, attractions, and tour guides.

Awards

1st Runner Up, ResearchSpark 3 Minute Thesis Competition

BUET

- Built BanglaHateCorpus (168K instances) and BanHateFormer — a Mixture-of-Experts Transformer achieving 91.33% F1 for low-resource NLP hate speech detection, 94× lighter than BanglaBERT.

April 2026

1st Runner Up, HULT Prize at BUET

BUET

- Co-developed ATLAS, a multimodal AI tutoring platform combining Vision Transformer OCR, Bayesian Knowledge Tracing, and BanglaBERT-based NLP grading for 24M underserved students.

February 2026

2nd Runners Up, Research Poster Presentation

BUET CSE Fest 2024

- Designed a real-time multimodal threat detection system fusing NLP-based rumor classification with CNN+RNN CCTV video analysis to cross-validate incidents across 16 threat categories.

October 2024